

LEC09 - Investing in Bonds and Mutual Funds

Chapter 12 - Investing in Bonds

Learning Objectives

1. Describe the characteristics of corporate bonds
 2. Discuss why corporations issue bonds
 3. Explain why investors purchase corporate bonds
 4. Discuss why federal, provincial, and municipal governments issue bonds and why investors purchase government bonds
 5. Evaluate bonds when making an investment
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Characteristics of Corporate Bonds

Bond: A corporation's written pledge to repay a specified amount of money along with interest

Face Value: The dollar amount that the bondholder will receive at the bonds maturity date

Maturity Date: The date on which the corporation is to repay the face value

Bond Indenture: The legal document that details all the conditions relating to a bond issue

Trustee: The bondholders representative

- Between time of issue and maturity, corporations pay interest to bondholders, usually semi-annually, at a stated rate (coupon rate)
 - *Dollar Amount of Interest = Face Value · Coupon Rate*
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Example

What is the annual interest amount for a \$1,000 bond that pays 3.75% interest?

$$\text{Dollar Amount of Interest} = 1000 \cdot 0.0375 = \$37.50$$

Why Corporations Issue Bonds

1. To borrow money to pay for major purchases
2. Difficult or impossible to sell stock
3. Finance ongoing business activities
4. To get money to operate or expand

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5. The interest paid to bondholders is a tax deductible business expense
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Types of Bonds

Debenture

- Backed by reputation of the company
- Most corporate bonds are debenture bonds

Mortgage Bond

- Secured by assets of company

Convertible Bond

- Can be exchanged by bondholder for common shares of the company
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Example

Jackson Metals, Inc., issued a \$1,000 convertible corporate bond. Each bond is convertible to 22 shares of the firm's common stock

1. What price must the common stock reach before an investor would consider converting their bond to common stock?
 - $= \frac{1000}{22} \approx \45.45
 2. If you owned a bond in Jackson Metals, would you convert your bond to a common stock if the stock's price reached the conversion price?
 - Depends on investment objectives, risk tolerance, relationship between market price of bonds and shares
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Why Investors Purchase Corporate Bonds

1. Interest Income
 - Investors know the interest rate
 - Interest will be paid to investors twice a year
 2. Appreciation of Bond Value
 - May be able to sell the bond to someone else at a higher price if the interest rate on the bond is higher than the market value
 3. Bond Repayment at Maturity
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Changes in Bond Value

- Price of corporate bonds may fluctuate until maturity

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- Caused by changes in interest rate
 - If interest rates fall, the price of the bond will increase since its coupon rate is higher (selling at a premium)
 - If interest rates rise, the price of the bond will decrease (selling at a discount)
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Government Bonds and Debt Securities

- Federal government sells bonds and securities to finance national debt and ongoing activities
 - Considered low risk investments
 - Offer lower interest rates than corporate bonds
 - Provincial governments issue bonds to fund program spending and fund deficits
 - Municipal bonds
 - Installment debentures
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Treasury Bills

Treasury Bill: Sometimes called a T-bill, is sold in a minimum unit of \$1,000 with additional increments of \$1,000 above the minimum

- T-bills with terms to maturity of 91, 186, or 364 days are currently auctioned on a bi-weekly basis, generally on Tuesday for delivery on Thursday
 - T-bills are discounted securities, and the actual purchase price you pay is less than the maturity value of the T-bill
 - The convention in Canada is to quote T-bills in yield
 - $Y = \frac{F-P}{P} \cdot \frac{365}{T}$
 - Y = Current yield for a T-bill
 - F = Face value of the T-bill
 - P = Purchase price
 - T = Term
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Example

Assume you purchased a 91 day T-bill with a purchase price of \$985 and a face value of \$1,000. What is the annualized rate of return?

$$Y = \frac{1000-985}{985} \cdot \frac{365}{91} \approx 6.11\%$$

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Bond Yield Calculations

Current Yield: The rate of return earned by an investor who holds a bond for a stated period of time

$$\text{Current Yield} = \frac{\text{Dollar amount of annual interest}}{\text{Current market value}}$$

Yield To Maturity: Takes into account the relationship among the bonds maturity value, the time to maturity, the current price, and the dollar amount of interest

$$\text{Yield to Maturity} = \frac{\text{Dollar amount of annual interest} + \frac{\text{Face value} - \text{Market value}}{\text{Number of periods}}}{\frac{\text{Market value} + \text{Face value}}{2}}$$

Example

Assume you purchased a corporate bond at its current market price of \$850 on January 1, 2023. It pays 5% interest annually on December 31, and will mature on December 31, 2032, at which time the corporation will pay you the face value of \$1,000

1. Determine the current yield on your bond investment at the time of purchase

$$\text{Current Yield} = \frac{1000 \cdot 0.05}{850} \approx 5.89\%$$

2. Determine the yield to maturity on your bond investment at the time of purchase

$$\text{Yield to Maturity} = \frac{50 + \frac{1000 - 850}{10}}{\frac{850 + 1000}{2}} \approx 7.03\%$$

Chapter 13 - Investing in Mutual Funds

Learning Objectives

1. Describe the characteristics of mutual fund investments
2. Classify mutual funds by investment objective
3. Evaluate funds for investment purposes
4. Describe how and why mutual funds are bought and sold

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Investing in Mutual Funds

- An investment chosen by people who pool their money to buy stocks, bonds, and other financial securities selected by professional managers who work for investment companies
- Each investor has a right to a proportional share of the assets of the fund and any income it earns
- Many people choose mutual funds for their retirement account

Professional Management: The people in charge of managing the fund

Diversification: Different asset classes (stocks, bonds, and other securities)

- Helps reduce risk by spreading risk across many sources

Investment Company: A firm that, for a management fee, invests the pooled funds of small investors in securities appropriate to its stated investment objectives

- Objective can be income, growth, etc.
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Characteristics of Mutual Funds

Net Asset Value: The current value of the securities contained in the mutual fund's portfolio minus the mutual fund's liabilities divided by the number of shares outstanding

- Calculated at the close of trading each day

$$\text{Net Asset Value} = \frac{\text{Value of the fund's portfolio} - \text{Liabilities}}{\text{Number of shares outstanding}}$$

Example

Given the following information, calculate the net asset value for the Altamira Bond mutual fund

- Total Assets – \$225,000,000
- Total Liabilities – \$5,000,000
- Total Number of Shares – 4,400,000

$$\text{Net Asset Value} = \frac{225000000 - 5000000}{4400000} = \$50$$

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Example

Given the following information, calculate the net asset value for the New Empire small-cap mutual fund

- Total Assets – \$350,000,000
- Total Liabilities – \$10,000,000
- Total Number of Shares – 17,000,000

$$\text{Net Asset Value} = \frac{350000000 - 10000000}{17000000} = \$20$$

Characteristics of Mutual Funds

- Load Fund
 - A mutual fund in which investors pay a commission every time they purchase or sell shares
 - Contingent deferred sales load → Pay upon withdrawal of funds
 - No-Load Fund
 - No sales charge
 - Front-end load
 - A sales fee charged with each purchase, reduces the funds invested
 - Back-end load
 - A sales fee charged with each sale
 - Also referred to as a contingent deferred sales load
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Example

Tong invested \$15,000 in the ADA Diversified Futures mutual fund. The fund charges a 5.5% commission when shares are purchased. Calculate the amount of commission Tong must pay

$$= 15000 \cdot 0.055 = \$825$$

Example

Matteo invested \$9,800 in the CI Harbour Growth and Income fund. The fund charges a 5.3% commission when shares are purchased. Calculate the amount of commission Matteo must pay

$$= 9800 \cdot 0.053 = \$519.40$$

Management Fees and Other Charges

Management Expense Ratio (MER): Expressed as a fixed percentage of the fund's total value

Special Fees

- Annual RRSP, RRIF, RESP trustee fee
 - Account set up fee
 - Short-term trading fee
 - Processing fee
 - Service fees
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Example

Christine invested a total of \$8,500 in the AIC Diversified Canada Mutual Fund. The management fee for this particular fund is 2.38% of the total investment amount. Calculate the management fee Christine must pay this year

$$= 8500 \cdot 0.0238 = \$202.30$$

Example

Mathew invested a total of \$52,000 in the New Colony Pacific Region mutual fund. The management fee for this particular fund is 0.80% of the total asset value. Calculate the management fee Mathew must pay this year

$$= 52000 \cdot 0.008 = \$416$$

Example

Over a four-year period, Pat purchased shares in the Barreau du Quebec Canadian Equity Fund as recorded in the table below

Year	Investment Amount	Price Per Share
2020	\$3,000	\$40
2021	\$3,000	\$50
2022	\$3,000	\$60

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2023	\$3,000	\$45
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1. At the end of the four years, what is the total amount invested?
 - $= 3000 \cdot 4 = \$12,000$
 2. At the end of the four years, what is the total number of mutual fund shares purchased?
 - $= \frac{3000}{40} + \frac{3000}{50} + \frac{3000}{60} + \frac{3000}{45} = 252 \text{ shares}$
 3. At the end of the four years, what is the average cost for each mutual fund share?
 - $= \frac{12000}{252} \approx \47.62
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Example

Assume that one year ago you bought 100 shares of a mutual fund for \$13.50 per share, you received a \$0.42 per-share capital gain distribution during the past 12 months, and the value of the fund is now \$17 per share. Ignoring tax, calculate the total return for this investment if you were to sell it now

Cost of investment	– \$1,350
Capital Gain Distribution	– $0.42 \cdot 100 = \$42$
Proceed of Disposition	– $17 \cdot 100 = \$1,700$
Gain on Investment	– $1700 + 42 - 1350 = \$392$
Return on Investment	– $\frac{392}{1350} = 29.04\%$